

# ELI — Memory

ELI v2.4.77

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## ELI Memory Subsystem

**Updated for v2.4.72.** Memory is governed by a storage policy (eli/memory/policy.py): every row has an origin, repeats are merged, weight follows a forgetting curve reinforced by use, and faded derived rows are archived. Recall applies a time window before ranking, and reports what it did. Turn retrieval is shared in eli/memory/retrieval.py; FAISS deletes use tombstones.

eli/memory/ holds 8,770 lines in 11 files: relational, full-text, vector and graph storage, all local SQLite and FAISS. Companion to project\_overview.md.

### The two layers

ELI’s memory is not “a dump of every interaction”. There are two layers:

1. **Conversation history** (conversation\_turns, conversations, session\_summaries): broad and chronological. A turn is kept unless the persistence gate (runtime/persistence\_gate.py) rejects it as junk. This is what a question about a period of time reads.
2. **Distilled memory** (memories, semantic, user\_patterns, the knowledge graph): a smaller set that the gate and the storage policy judged worth keeping, labelled by origin, merged when repeated, and faded when nothing reinforces it. Faded derived rows move to memories\_archive instead of being deleted.

The memory-runtime report states this explicitly (EXPLAIN\_MEMORY\_RUNTIME, memory internals), with live counts for each layer.

## Files

| File                 | Lines | Role                                                                                            |
|----------------------|-------|-------------------------------------------------------------------------------------------------|
| memory.py            | 5,677 | the Memory class (69 public methods), schema, DBPaths, upkeep, module facade                    |
| policy.py            | 118   | pure storage-policy functions (origin, dedupe key, strength, archive, merge)                    |
| retrieval.py         | 253   | shared turn retrieval (retrieve_for_turn), turn cache, time window                              |
| claims.py            | 266   | dated claims about the user: valid-from and valid-to, learned-at, supersession                  |
| unified_retrieval.py | 168   | the orchestrator stages consume retrieve_for_turn through it; formats the verified-memory block |
| vector_store.py      | 637   | FAISS index, embedder, tombstones                                                               |
| knowledge_graph.py   | 643   | entity and relation graph                                                                       |
| habits_memory_db.py  | 466   | habit and legacy memory helpers                                                                 |
| system_index.py      | 278   | indexed apps, executables, files                                                                |
| memory_truth.py      | 188   | read-only inspection used by status surfaces                                                    |
| memory_adapter.py    | 131   | compatibility adapter                                                                           |
| __init__.py          | 278   | facade (get_memory, get_agent_memory, path helpers)                                             |

## The storage policy (policy.py)

Every stored row gets an **origin** from `classify_origin(source, kind, tags, text)`:

| Origin    | Meaning                                            | Factor on half-life |
|-----------|----------------------------------------------------|---------------------|
| user_said | the owner said it or confirmed it                  | 1.0                 |
| eli_said  | ELI's own assistant text                           | 0.5                 |
| tool      | executor or observation output                     | 0.35                |
| news      | news digests and reflections                       | 0.25                |
| telemetry | bookkeeping ( <code>is_bookkeeping_memory</code> ) | 0.15                |

Rules, all in `policy.py` and pinned by `tests/test_memory_policy.py`:

- **Dedupe.** `text_key` is a sha1 of the normalised text (first 20 hex characters). A repeat bumps `seen_count` and `last_seen` instead of adding a row; `merge_group` keeps the earliest event time.
- **Strength.**  $2^{**} (-days\_since\_last\_touch / half\_life)$ , floored at `MIN_WEIGHT` (0.05). The half-life is the base half-life  $\times$  origin factor  $\times (1 + 3 \times importance) \times (1 + 0.6 \times \ln(1 + uses))$ , where `uses` are repeat sightings plus recalls.
- **Base half-life.** 30 days, scaled by the square root of the median gap between days the owner used ELI, clamped to 14-120 days (`adaptive_half_life_days`).

- **Pinning.** A user\_said row with importance  $\geq 0.85$  stays at strength 1.0.
- **Archiving.** should\_archive is true only for a derived (not user\_said) row that has never been recalled, has importance  $< 0.85$ , and whose weight is  $\leq 0.12$ .
- **Vector index.** Telemetry rows are not embedded (wants\_vector\_index).
- **Evidence about the user.** Only user\_said counts (counts\_as\_evidence\_about\_user).

Columns added to memories: event\_ts, seen\_count, last\_seen, last\_recalled, recall\_count, origin, text\_key. Tables added: memories\_archive, memory\_meta. The semantic table gained evidence\_count, last\_seen, evidence.

### What counts as the same fact, and what strengthens a memory

- **Dedupe key.** normalise\_text lower-cases, single-spaces and trims sentence punctuation from the edge of each word, and keeps meaningful marks. “I love C++”, “I love C#” and “I love C” are three facts, “balance -5” is not “balance 5”, and “3.5” is not “35”. “Not fine!” and “not fine” are one. A key-version marker makes upkeep recompute old keys once (rekey\_text\_keys); it never merges.
- **Retrieval is exposure, not use.** Recalling a memory only increments exposure\_count. Its strength is reinforced by record\_recall\_outcome(ids, helped=True), which the engine calls when the next user message does not correct the answer those memories went into; a correction lowers the weight and increments corrected\_count. A wrong memory that keeps being retrieved cannot make itself stronger. recall\_count now means answers a memory helped with, and only that protects a row from archiving.
- **What enters durable memory.** persistence\_gate.should\_store\_memory\_text rejects filler and reactions (“haha”, “lol that is funny”, “hello there”), text under four words, questions, and ELI’s own internal output. Conversation turns are still logged in full; only durable memory is filtered.

### Dated claims (claims.py)

memory\_claims holds what ELI believes about the user as bitemporal rows: valid\_from/valid\_to (when it was true) and recorded\_at/superseded\_at (when ELI knew it). A first-person statement such as “I work nights now”, “I moved to Berlin”, “I work at Acme Corp” or “my dog is called Max” becomes a claim when it is stored. A single-valued relation supersedes the old value and keeps it as history, “I used to work nights” is recorded as history only, “I no longer work nights” retires the standing claim, and the same value again confirms instead of duplicating. Memory.claims\_current(), claims\_during(start, end), claims\_known\_at(when) and claim\_history(relation) answer “what is true now”, “what held in spring” and “what did you believe on 1 March”. Retrieval adds the relevant claims to the evidence (a period question gets what held then), and deleting the source memory withdraws its claims. Extraction is a small exact set of patterns, not a model call.

### Forgetting, lineage and conflicts

- **Forget.** “forget that my locker is 212” lists the matching memories and asks; on yes (or confirm forget memories 12 15) Memory.forget(ids) deletes them and everything derived from them: derived rows through memory\_lineage, index entries and vector tombstones, claims, recall log rows, semantic-fact quotes, session summaries that carry the same content, profile items and briefs, stances and belief revisions, graph entities created with the memory, and the conversation turns. It returns a report of what was removed.
- **Lineage.** link\_derivation(child, parents) records that a summary or reflection was made from other memories; lineage\_root and independent\_sources mean a fact and its summaries count as one source. A claim’s confirmations count distinct roots only.
- **Conflicts.** A claim that is not the user’s own word and contradicts one that is stands beside it as disputed (both marked contested) instead of replacing it. claim\_conflicts() lists them, claims.conflict\_question phrases the question, and resolve\_claim\_conflict records the answer. Hedged statements (“maybe”, “thinking of”, “if I”) are never claims.
- **Retrieval scope.** retrieve\_for\_turn returns searched: live memory, exact codes and quoted phrases (looked up as written, so an embedding cannot blur “ORCHID-7319”), dated claims, and the

archive, which is searched only when live evidence is thin and its hits are labelled. The diagnostics block states that scope, so “nothing stored” is only claimed for what was searched.

## Daily upkeep

`Memory.run_upkeep()` runs at most once a day (`upkeep_due()`), started asynchronously at engine start and after responses (`upkeep_async()`), and skipped under `ELI_TEST_MODE`. Steps:

1. **backfill**: label existing rows with origin, key and event time (after copying the database to a `.pre_policy.bak` file the first time). A dry run reports what it would fill and writes nothing; 1a. **rekey**: recompute dedupe keys once after the key definition changes;
2. **consolidate**: merge exact duplicates and rebuild `memories_fts`;
3. **decay**: recompute weight with the policy strength (idempotent: a pure function of age, importance and use, so a missed run cannot leave a row over-weighted);
4. **archive**: move faded derived rows to `memories_archive` (`search_archive`, `restore_from_archive` bring them back);
5. **identity**: merge legacy user ids into one stable owner id (`owner_id`, kept in `memory_meta`); only UUID-shaped legacy ids are merged;
6. **learning**: remove repeated identical belief revisions, cap corroboration at the number of days the owner has used ELI, and drop ledger copies of replay rows;
7. **semantic**: fold repeated facts into one row with `evidence_count` and up to five quotes.

A run only marks the day done when every step succeeded; otherwise `run_upkeep` reports status: partial with the failed steps, and upkeep retries after an hour. `integrity_report()` reports the state of all of this for the health checks: ok is false when vectors are missing, and status is healthy or degraded.

## The Memory class (`memory.py`)

A single class that owns most persistent concerns:

- **Memory**: `store_memory`, `add_memory`, `recall_memory`, `search_memory`, `search_memories`, `memories_between`, `get_recent_memories`, `adjust_weight`, `apply_weight_decay`, `consolidate_memories`, `archive_faded`, `search_archive`, `restore_from_archive`, `store_episodic`, `store_semantic`, `recall_semantic`, `store_reflective`.
- **Conversation**: `add_conversation_turn`, `store_conversation`, `get_conversation_history`, `get_recent_conversations` (with `since`, `until`, `role`), `get_recent_turns_since`, `search_conversations`, `get_turns_for_day`, `save_session_summary`, `get_session_summaries`.
- **Habits**: `log_habit_event`, `get_habit_events`, `add_habit_rule`, `get_habit_rules`, `get_detected_habits`, `record_habit_run`.
- **Self-improvement and learning**: `log_learning_event`, `log_failure`, `log_correction`, `add_observation`, `log_improvement`, `add_capability_proposal`, `propose_capability`, `get_pending_proposals`, `get_recent_failures`, `get_recent_improvements`, `get_recent_observations`. These tables are written on the **agent** database (`SelfImprovementEngine` opens `get_agent_memory()`); the user database holds empty copies.
- **Health**: `get_stats`, `get_dashboard_counts`, `get_db_routing_info`, `integrity_report`.

`vector_store` is a lazy property; the knowledge graph is reached through `knowledge_graph.get_knowledge_graph()`.

## Schema

The blank `user.sqlite3` template holds 27 tables and three FTS5 indexes. `memories` (+ `memories_fts`), `memories_archive`, `memory_meta`, `semantic`, `conversation_turns`, `conversations`, `session_summaries`, `kg_entities` (+ FTS), `kg_relations`, `recall_log`, `runtime_events`, `learning_replay`, `observations`, `habits`, `habit_events`, `habit_rules`, `user_patterns`, `user_model`, `eli_stances`, `belief_revisions`, `corrections`, `failures`, `error_tracking`, `improvements`, `capability_proposals`, `news_articles` (+ FTS), `news_reflections`. `agent.sqlite3` holds `agent_dispatches`, `agent_metrics` and the self-improvement tables.

emotion\_events is owned by cognition/emotion\_timeline.py, not the Memory class: it opens user.sqlite3 directly and creates its table on first use. Columns: ts, user\_id, session\_id, detected, expressed, valence (negative, positive or neutral), confidence, source, arousal, user\_text and eli\_prior\_action (the action ELI ran on the preceding turn, so it can ask whether the mood turned because of something it did). Reads come back ORDER BY ts DESC, id DESC; the id tiebreak matters because several reads can land within one second.

Each conversation turn is stored in conversation\_turns. A learning\_replay row is also written per turn (it carries action, outcome and reward), and the evidence ledger records the turn once in runtime\_events; replay rows are not mirrored into the ledger.

## Shared turn retrieval (retrieval.py)

The single owner of semantic and conversation recall for a turn. BusMemoryAgent and the orchestrator both call retrieve\_for\_turn(), so a query is not searched twice with different budgets.

- **Turn cache:** 8 seconds per process. The key includes session, user, query, time window and every limit (semantic\_limit, conv\_limit, recent\_limit, summary\_limit, hop2\_limit, merge\_cap, verified\_only), so a deeper pass never gets the shallower result. invalidate\_turn\_cache() clears it.
- **Time window first.** When the planner finds a period in the question (query\_planner.parse\_window), memories dated inside it are fetched by date (Memory.memories\_between, ordered by importance × weight) and merged with the topic hits, hits outside the window are dropped, and the user's turns inside the window are read from conversation\_turns. The result carries window\_stats (candidates, added by date, in window, turns) and the orchestrator logs and reports it (memory\_diag).
- **Only questions get a window.** plan\_window applies a period only to a question or recall request, so a statement that says “last night” or “today” cannot narrow retrieval to that period.
- **Explicit dates.** parse\_window reads ISO dates (“2026-03-03”), “3 March”, “March 3rd 2025”, ranges (“between 1 May and 10 May”), “in June 2025” and “during 2024” as well as relative periods. A date with no year is its latest past occurrence.
- **Token budget.** context\_budget.chars\_per\_token() measures the loaded model's own tokenizer and falls back to 3.5 characters per token.
- **Hop-2 deepening** when the first hits are sparse; **heuristic rerank** via rerank\_candidates(); **contradiction detection** through the bus.
- **Dated evidence.** Each recalled line carries its event date (evidence\_format); working memory keeps the original date rather than the date it was re-pinned.

## recall\_memory: the hybrid retriever

The primitive retrieve\_for\_turn() builds on:

1. **FAISS and FTS5 both run** and are fused by reciprocal rank (fuse\_ranked\_lists); a LIKE scan covers the case where FTS returns nothing. keyword\_only=True skips FAISS.
2. **Noise filtering** (memory\_exclusion\_sql in core/self\_provenance.py): ELI's own bookkeeping kinds and sources never resurface as recalled user memories.
3. **Importance-weighted ordering** via COALESCE(importance, 0.5).
4. **Reinforcement.** A recalled row gets last\_recalled and recall\_count updated and its weight reset to 1.0; the recall is written to recall\_log with its memory\_id.

Column detection (\_memory\_table\_columns) guards against schema drift between versions.

## Vector store (vector\_store.py)

FAISS IndexFlat, embeddings from a local nomic embedder (llama.cpp).

- \_embed\_lock (RLock) serialises embedding: the embedder is not thread-safe, which is why orchestrator retrieval is sequential.
- Metadata lives in meta.json (migrated from a legacy meta.pkl).

- Singleton via `get_vector_store()`; shutdown-aware; `reset_vector_store()` for rebuilds.
- **Tombstones:** `mark_memory_deleted(id)` writes to `meta.tombstones.json` next to the metadata; search skips tombstoned rows without a rebuild; `compact_tombstones(live_ids)` reclaims space.
- `store_memory()` returns `vector_indexed: bool`, a failed index write logs at warning level, and `fire_memory_uncertainty_event()` drives the World tab's `memory_uncertainty` bar.
- Tombstoning is a per-call-site discipline: there is no shared `delete_memory_row()` helper, so any direct `DELETE FROM memories` must tombstone the vector itself (`consolidate_memories` and `profile_extractor._scrub_onboarding_snapshot` do).

## Knowledge graph (`knowledge_graph.py`)

`kg_entities(name, type, aliases, description, confidence)` and `kg_relations(subject_id, predicate, object_id, weight, source)`: a subject-predicate-object graph. FTS5 over entities with insert, update and delete triggers for fuzzy search `entities`. `upsert_entity`; `context_for_prompt` gives lightweight SQLite-only prompt context with no embedding. A stop-word list stops common words becoming entities.

## Truth layer (`memory_truth.py`)

`inspect_sqlite` and `inspect_vector_store`: read-only inspection used by status surfaces; reads `vectors/meta.json` and falls back to the legacy pickle. Backs `truth_report` and the memory-status surfaces.

## Promotion across tiers

Promotion is a fan-out from `user_patterns`:

```
turns → user_patterns ─┬→ semantic (profile_extractor._promote_to_semantic)
                       └→ KG entities and relations
                           (persona_updater._populate_kg_from_user_patterns)
```

`_promote_to_semantic` applies a durable/transient prefix filter, dedupes against semantic and fires only when `_insert_user_pattern` actually inserted, so reaffirmations do not re-promote. Corroboration counts days the owner reaffirmed a fact, not extraction passes, and a passing mention cannot overwrite a fact the owner stated and reaffirmed (`_insert_user_pattern` weighs provenance and corroboration; each supersession is recorded in `belief_revisions`, and repeats are deduplicated). `store_episodic` and `store_semantic` are thin aliases and are not the promotion path.

## Measuring it

`tools/eval/memory_bench.py` runs 11 deterministic scenarios against a temporary database, with no model: extraction, multi-session recall, a relative and an explicit time window, knowledge update, abstention, provenance, deletion, no destructive merge, no self-reinforcement, and chatter not becoming durable knowledge. They follow the LongMemEval ability categories but are written for ELI; they are not the official dataset. `tests/test_memory_bench.py` runs them, so a memory change that regresses one fails the suite.

## Honest assessment

- **Strong:** genuinely hybrid (vector, FTS5, graph) on one local SQLite foundation; a measurable storage policy with an audit trail; recall that filters by date before ranking; noise filtering that keeps ELI's own reflections out of recall; embedder serialisation that avoids the segfault.
- **Weak:**
  1. `memory.py` is a 5.7k-line class spanning unrelated concerns (semantic, conversation, habits, learning, failures, capabilities, upkeep). It wants splitting along those seams.
  2. **Schema sprawl.** `memories` and a legacy memory, `conversations` and `conversation_turns`, plus a standalone semantic table; inline column detection compensates for schema churn.

3. **Narrow intake.** Promotion works, but it is fed by regex extraction plus one LLM pass per session, so `user_patterns`, the semantic tier and the knowledge graph stay small relative to the conversation history. Widening the extractor is the work.
4. **Retrieval cannot detect nonsense.** Similarity bands for genuine and gibberish queries overlap, so no absolute similarity floor separates them; the relative cutoff in `vector_store` tightens the candidate pool but is not a relevance oracle.
5. **No shared delete-and-tombstone helper** (above).

`eli/cognition/reranker.py`'s heuristic reranker weighs overlap 0.45, importance 0.20, weight 0.15, recency 0.10. Importance and recency alone (0.45 combined) could match the overlap term's full weight, so a memory saved with "please remember this" (importance near 1.0) could outrank a memory that actually matches a vague query, on nothing but being generally important and recent — proven against a live database, where a stored test code beat memories that were actually about the asked-for week. The importance/weight/recency term is now gated: full credit only above a real overlap floor (0.2), a reduced share when a retriever found the candidate at all but overlap is weak or incidental (a single shared word in a long memory no longer counts as "on topic"), and the smallest share when neither applies. `tests/test_reranker_importance_does_not_override_relevance.py` pins this.

## Recent-history window

The user-facing tunable `cog.mem_recent_turns` (Settings ► Cognition; default 24, maximum 80) controls how many recent turns enter the prompt. `RECENT_HISTORY_CAP` no longer exists (the old memory-evidence module was removed); the prompt assembler and `context_budget` do the budgeting, and a recall question keeps a floor of a third of the window for memory context.